

AWS CloudWatch Lab: Monitor Amazon EC2 Instance Metrics

Summary

In this lab, **Amazon CloudWatch** was used to monitor the CPU utilization of an **Amazon EC2** instance. A new EC2 instance was launched using a User Data script that generated CPU load, allowing CloudWatch to collect and display performance metrics. The lab explored how CloudWatch collects metrics at different intervals, how to view instance metrics, and how to monitor CPU utilization over time. Finally, the monitoring results were verified using CloudWatch metrics, and the importance of terminating unused EC2 instances to avoid charges was highlighted.

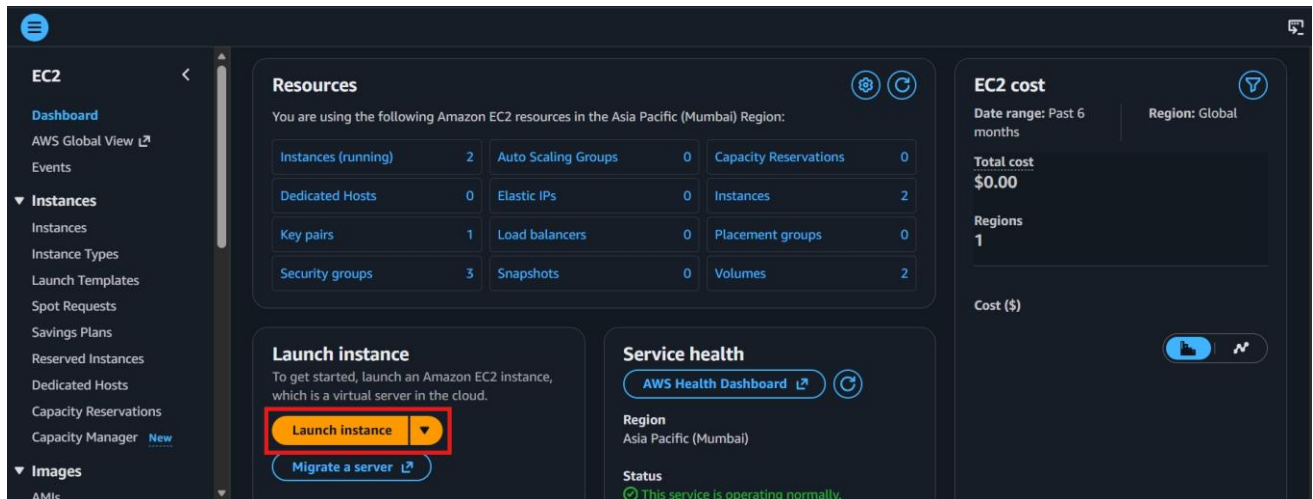
Let's Start:-

- **Amazon CloudWatch** is a monitoring service used to collect and visualize metrics from AWS resources.
- By default, **Basic Monitoring** collects EC2 metrics every **5 minutes**.
- **Detailed Monitoring** collects metrics every **1 minute** and incurs additional charges.
- This lab monitors **CPU Utilization** of an EC2 instance using CloudWatch Metrics.

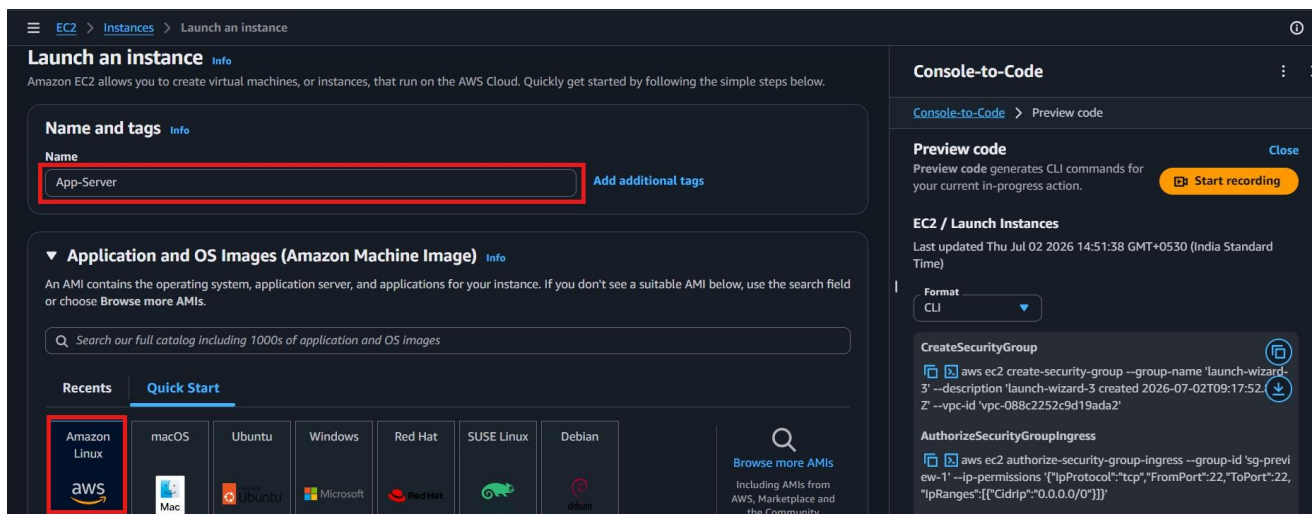
Lab Steps

Step 1 – Launch a New EC2 Instance

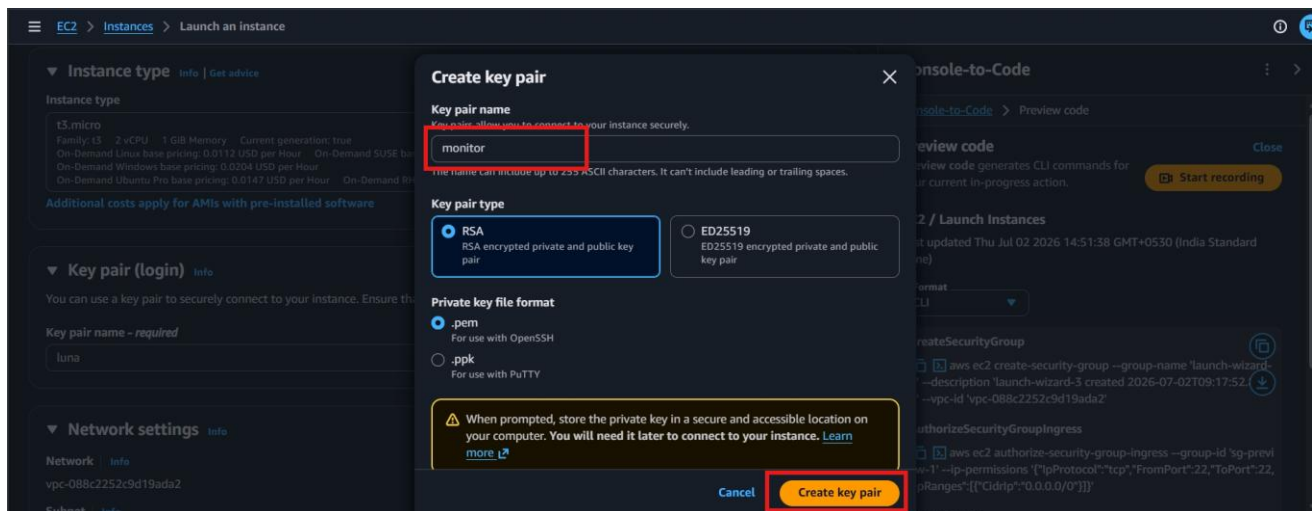
- Sign in to the **AWS Management Console**.
- Navigate to **Amazon EC2**.
- Click **Launch Instance**.



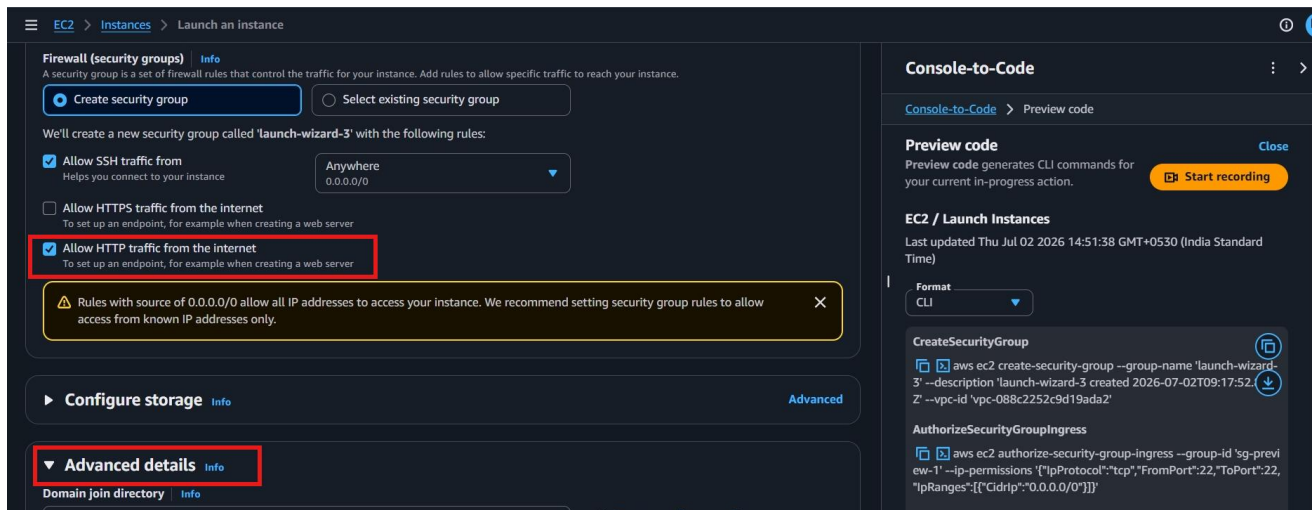
- Enter the instance name as **New Server**.
- Select **Amazon Linux** as the AMI.



- Choose a **Free Tier eligible** instance type (**t2.micro** or **t3.micro**).
- Select an existing or new **Key Pair**.

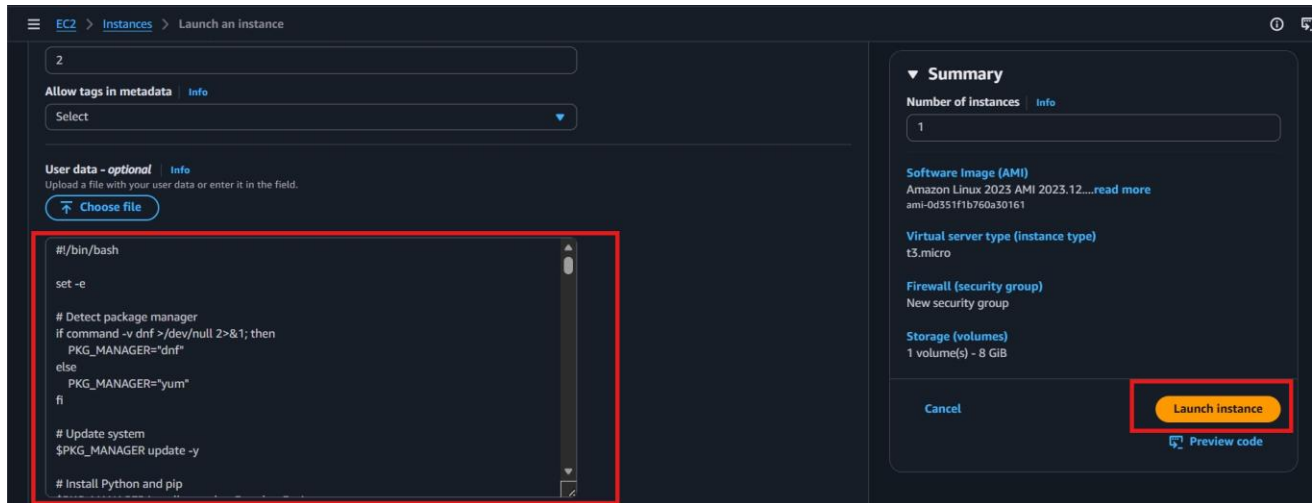


- Use the **Default VPC**.
- Enable **Allow HTTP Traffic**.
- Scroll to **Advanced Details**.



Step 2 – Add the User Data Script

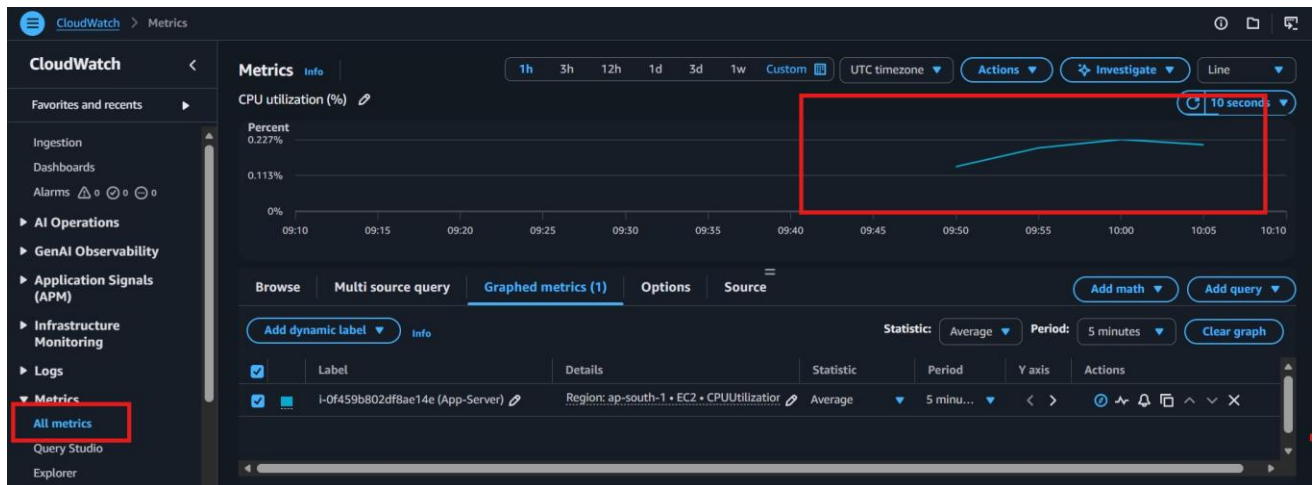
- Open the provided **User Data Script (CPUcontrol.sh)** file from the lab resources
- Paste the script into the **User Data** section.



- Click **Launch Instance**.
- Wait until the EC2 instance reaches the **Running** state.

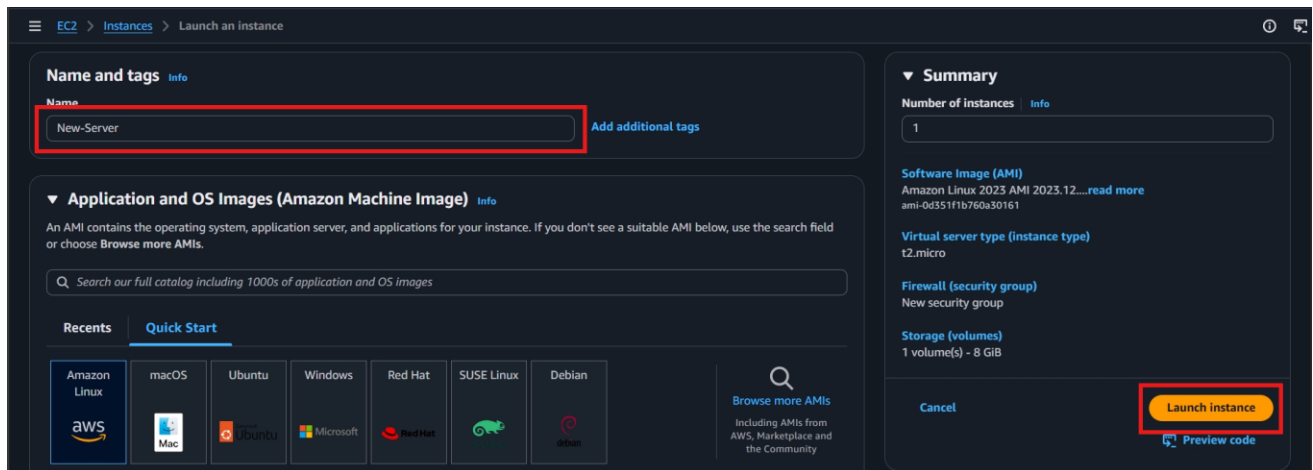
Step 3 – Open Amazon CloudWatch

- Navigate to **Amazon CloudWatch**.
- Select **Metrics** from the left navigation pane.
- Open the **EC2** namespace.
- Select **Per-Instance Metrics**.

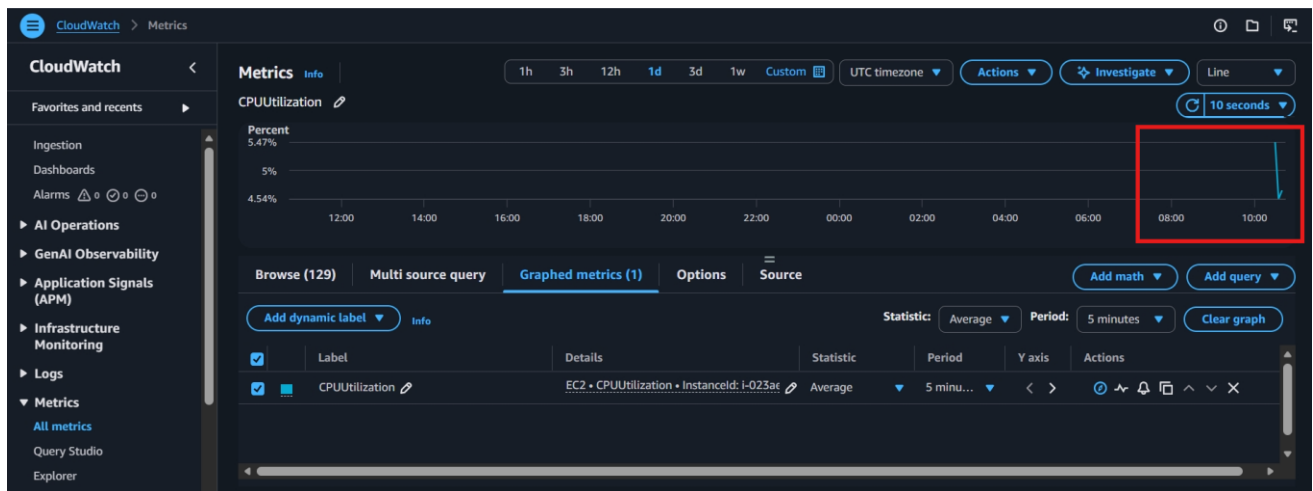


Step 4 – Monitor the New EC2 Instance

- Return to the **Amazon EC2** console.
- Copy the **Instance ID** of the newly created instance – follow the same process(use key-pair → monitor).



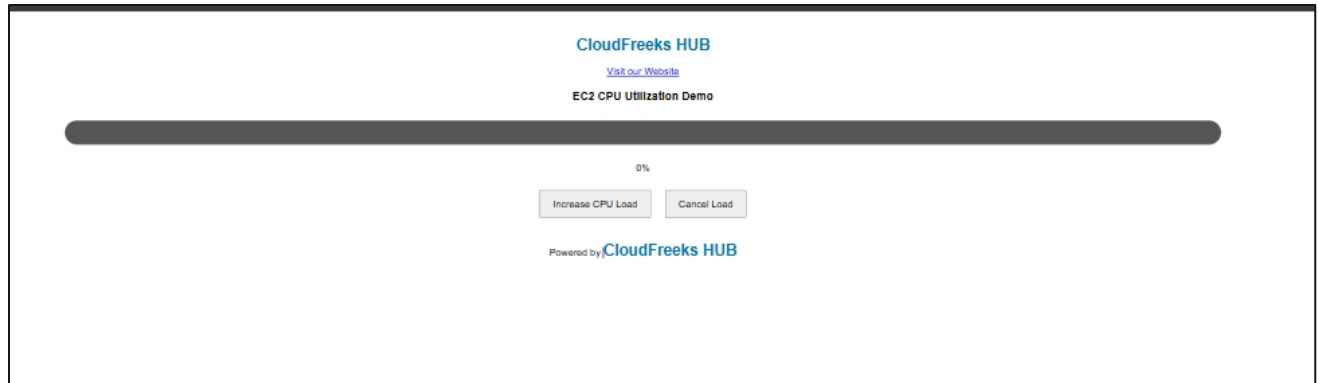
- Search for the instance inside **Per-Instance Metrics**.
- Select the **CPUUtilization** metric.



- Remove any previously selected instance metrics if required.

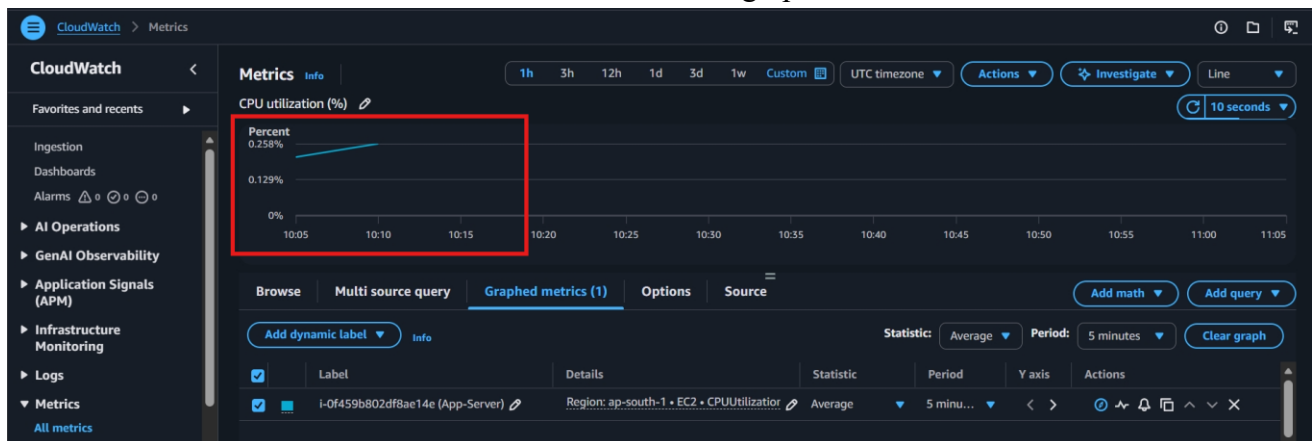
Step 5 – Access the Web Application

- Return to the **Amazon EC2** console.
- Copy the **Public IPv4 Address** of the instance.
- Open a browser and enter:
- **http://**
- Confirm that the web application loads successfully.



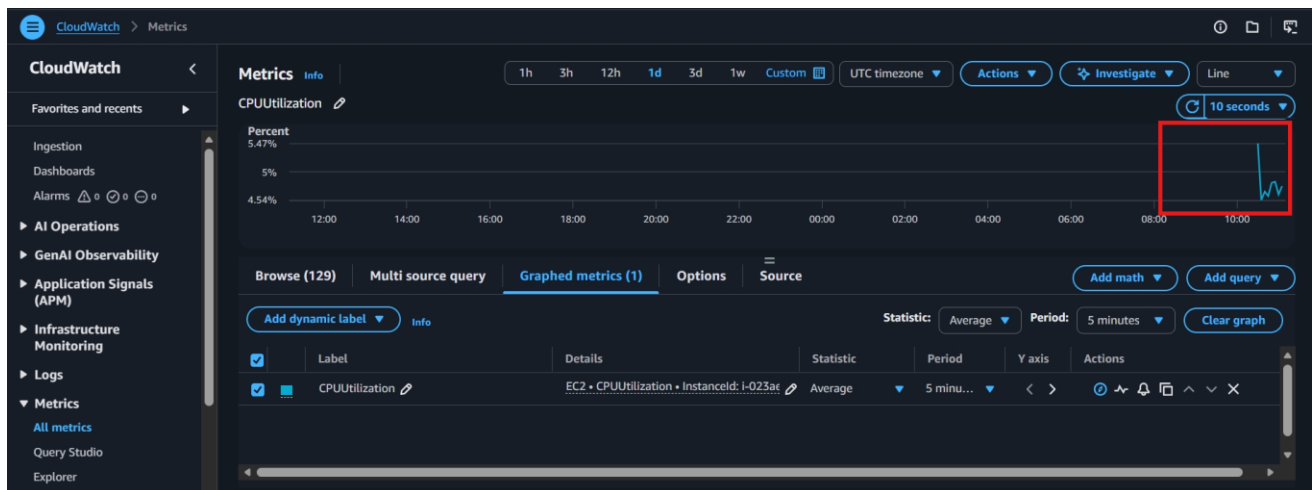
Step 6 – Generate CPU Load

- Click the option to **Increase CPU Load**.
- Observe that the CPU utilization begins to increase.
- Return to **Amazon CloudWatch** and monitor the graph.



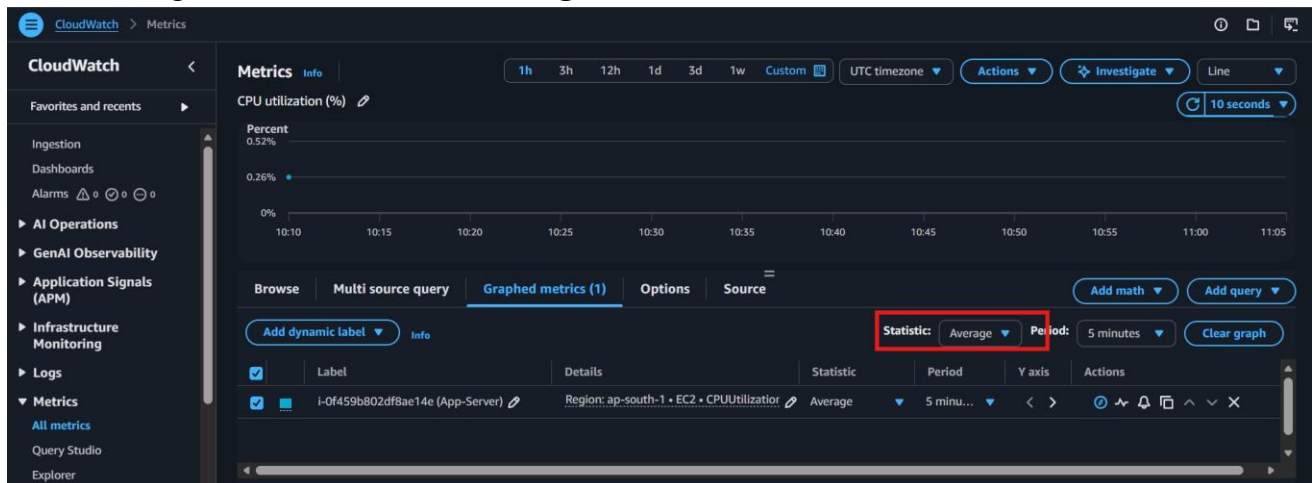
Step 7 – Stop the CPU Load

- Return to the application.
- Click **Cancel Load**.
- Wait a few minutes.
- Observe the CPU utilization decreasing in the CloudWatch graph.

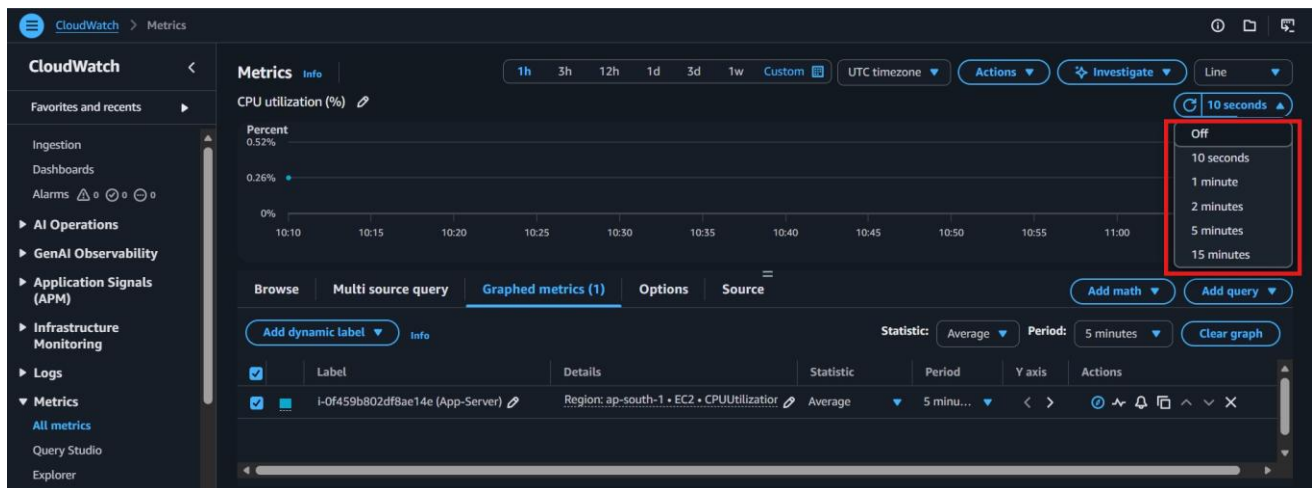


Step 8 – Explore CloudWatch Metrics

- Change the **Statistic** to view **Average**, **Minimum**, and **Maximum** CPU utilization.

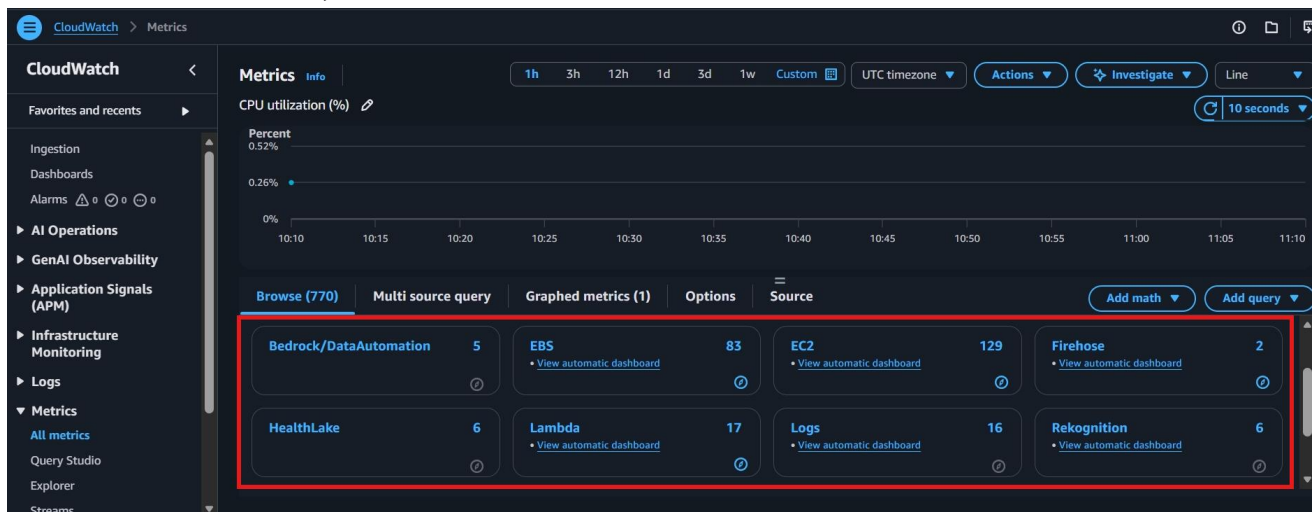


- Switch to the **Data Table** view to observe collected metric values.
- Modify the **Time Range** (1 Hour, 3 Hours, etc.) to analyze historical data.
- Observe that **Basic Monitoring** collects one metric every **5 minutes**.



Step 9 – Explore CloudWatch Namespaces

- Return to **Metrics**.
- Observe the available **Namespaces**.
- Open the **EC2** namespace.
- Note that CloudWatch can also monitor services such as **Amazon S3**, **AWS Lambda**, **Amazon Kinesis**, and **Elastic Load Balancer** when those resources exist.



Verification

- The EC2 instance was successfully launched.
- The User Data script generated CPU load successfully.
- **Amazon CloudWatch** displayed CPU utilization metrics for the EC2 instance.
- Average, Minimum, and Maximum CPU utilization metrics were successfully viewed.
- The CPU utilization graph reflected increases and decreases in workload.
- CloudWatch namespaces were explored successfully.
- The EC2 instance was terminated after completing the lab to avoid unnecessary AWS charges.